Flight controls — slat actuation

Service bulletin MRDX-27-1042 — Replacement of the slat feedback position pick-off units

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Responsible partner	AERALIS S.A.S. (U8025)	**Originator**	AERALIS S.A.S. — Customer Services Valmont (U8025)
Security	02 — Restricted	**Quality assurance**	Second verification (tabtop)
Applicability	MRDX-42, MSN 0021 to 0184; MRDX-46, MSN 0201 to 0233		

Service bulletin number	MRDX-27-1042
ATA chapter	27-81-00
Compliance category	sbcc02
Task category	sbtc03
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REVISION INFORMATION

Revision 2 adds the MRDX-46 aircraft to the effectivity and corrects the man-hour estimate of the accomplishment instructions from 6.0 to 7.5 man-hours. The material information is unchanged.

SUMMARY

This service bulletin gives the instructions to replace the two feedback position pick-off units (FPPU) of the slat actuation power control unit with an improved standard. The improved units have a sealed bearing housing that prevents the ingress of moisture which, on the units in service, has caused an increased drift of the position signal and the fault message `SLAT SYS 1 FAULT` in service.

1. PLANNING INFORMATION

A. Effectivity

This service bulletin applies to the aircraft listed below that do not have modification 27-8117 embodied in production.

MODEL	SERIAL NUMBERS
MRDX-42	MRDX-420021 to 0184, except 0102, 0117 and 0140
MRDX-46	MRDX-460201 to 0233

B. Reason

Nineteen occurrences of slat position signal drift have been reported in service between January 2024 and September 2025. In each case the moisture that entered the bearing housing of the feedback position pick-off unit caused a slow degradation of the resolver output. The condition is detected by the monitoring function before it affects the surface position, but it causes an unscheduled removal and, in seven of the reported cases, a departure delay.

C. Description

The two feedback position pick-off units are removed from the power control unit and replaced by units of part number 27DA1310000000. The new units are dimensionally and electrically interchangeable with the units they replace. The wiring, the software standard of the slat flap control computers and the rigging of the transmission are not affected.

The accomplishment of this service bulletin embodies modification 27-8117 on the aircraft.

D. Compliance

Recommended. Aeralis recommends that this service bulletin is accomplished at the next scheduled maintenance check that gives access to the belly fairing, and in any case within 24 months of the date of this revision.

E. Approval

The technical content of this service bulletin is approved under the authority of EASA design organization approval EASA.21J.0000.

F. Manpower

The estimated man-hours are 7.5 man-hours for one aircraft, which does not include the time to gain access to and close the belly fairing. The estimate is based on two technicians working at the same time.

G. Weight balance

The change of mass is negligible. The moment change is inside the tolerance of the weight and balance manual and no revision of the aircraft weighing record is required.

H. Electrical load

Not affected.

I. Interchangeability

The units are two-way interchangeable at aircraft level. A power control unit that has one unit of each standard installed is not permitted.

J. Publications affected

The aircraft maintenance manual, the illustrated parts catalogue and the trouble shooting manual are revised at the next scheduled revision.

2. MATERIAL INFORMATION

A. Material price

The material is supplied as kit 27-1042-01 at no charge for the aircraft in the effectivity of this service bulletin, on receipt of an order placed before 31 December 2026.

B. Material kit

Table 1 Kit 27-1042-01 contents, one kit per aircraft

Part number	Designation	Qty	Replaces
27DA1310000000	Pick-off unit, feedback position, sealed	2	27DA1300000000
NSA8102-118	Packing, preformed	4	—
NSA935403-08	Lockwire, 0.8 mm, 1 m coil	1	—

C. Material availability

The kit is available from the Aeralis spares centre with a lead time of six weeks. The units that are removed are to be returned within 60 days.

3. ACCOMPLISHMENT INSTRUCTIONS

A. General

> **WARNING**
>
> Make sure that the green and the yellow hydraulic systems are depressurized and that the slat travel area is clear before you start this procedure.

Do the removal and installation with the power control unit installed on the aircraft. It is not necessary to remove the power control unit.

B. Procedure

1. Open, safety and tag the circuit breakers 4027GA and 4027GB on panel 121VU.
2. Open the fwd belly fairing access panel 196AB.
3. Disconnect the electrical connector of the first feedback position pick-off unit, remove the four attachment screws and remove the unit.
4. Install a new packing and the new unit 27DA1310000000. Tighten the four screws to 6 Nm to 8 Nm and install the lockwire.
5. Do steps 3 and 4 again for the second unit.
6. Remove the safety tags, close the circuit breakers and do the slat system operational test.
7. Record the embodiment of modification 27-8117 in the aircraft technical log and in the modification status record.

4. ADDITIONAL INFORMATION

A. Para

Report the accomplishment of this service bulletin through the customer services portal within 30 days. Quote the aircraft serial number, the date of accomplishment and the serial numbers of the units that were removed.